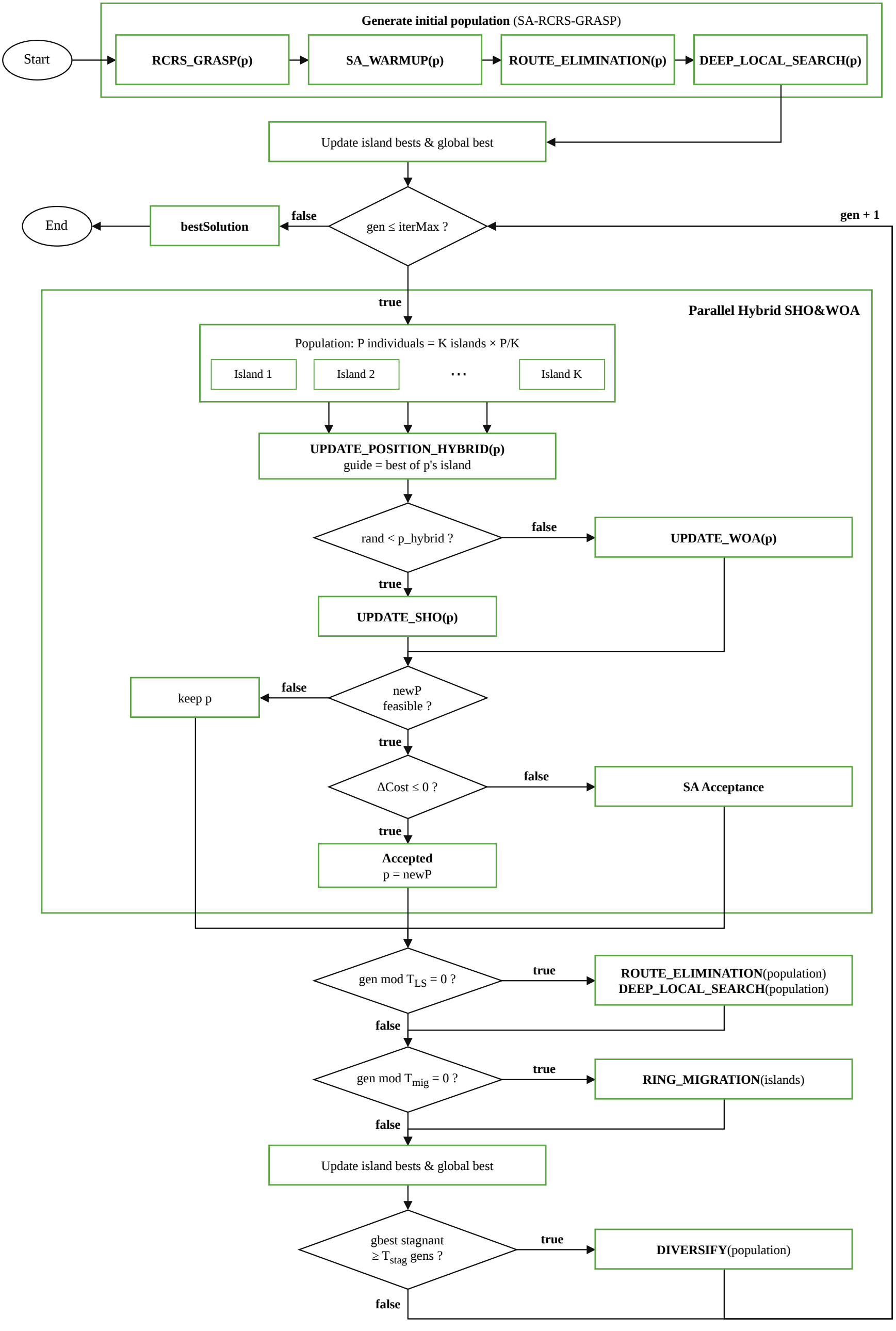
Start
Generate initial population (SA-RCRS-GRASP)
RCRS_GRASP(p)
SA_WARMUP(p)
ROUTE_ELIMINATION(p)
DEEP_LOCAL_SEARCH(p)
Update island bests & global best
gen ≤ iterMax ?
false
bestSolution
End
gen + 1
true
Parallel Hybrid SHO&WOA
Population: P individuals = K islands × P/K
Island 1
Island 2
...
Island K
UPDATE_POSITION_HYBRID(p)
guide = best of p's island
rand < p_hybrid ?
false
UPDATE_WOA(p)
true
UPDATE_SHO(p)
newP feasible ?
false
keep p
true
ΔCost ≤ 0 ?
false
SA Acceptance
true
Accepted
p = newP
gen mod T_LS = 0 ?
true
ROUTE_ELIMINATION(population)
DEEP_LOCAL_SEARCH(population)
false
gen mod T_mig = 0 ?
true
RING_MIGRATION(islands)
false
Update island bests & global best
gbest stagnant ≥ T_stag gens ?
true
DIVERSIFY(population)
false